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AI SECURITY

AI security engineer working on the red-teaming and evaluation of LLM and agentic systems, with a defensive background in SOC telemetry and detection engineering. CompTIA CySA+ certified. Current focus: the reliability of the automated judges the jailbreak field uses to score attacks — and what those scores hide. Comfortable owning the full path from telemetry pipeline to trained model to the adversarial evaluation of both.

RESEARCH

Judge-Shaped, Not Harm-Shaped

preprint in preparation | github.com/secslayer/jlens-fuzz

Measurement validity in jailbreak success judges; activation-guided fuzzing.

- Found the standard **hubert233/GPTFuzz** judge — reused across the jailbreak literature without re-validation — inflates attack success by rewarding jailbreak-*shaped* persona/roleplay framing over actual harmful content.
- Built a stricter two-stage replacement (deterministic refusal pre-filter + anti-roleplay LLM-judge rubric); reported honestly that it *still* passed a completion that refuses and offers crisis-support resources — the failure mode survives a targeted fix.
- Secondary result reported as an honest null: activation-guided span mutation (refusal direction, difference-in-means) shows no consistent ASR advantage over uniform mutation on either target, with the mechanism independently confirmed engaged rather than silently inactive. Smoke-scale (n=5/condition) under a hard free-tier compute budget, stated plainly.
- Full provenance discipline: every reported number traced to a committed **results/*.json**; CI content-scanning enforces that no raw jailbreak strings ship.

AJF reproduction

github.com/secslayer/reproductions

- Independent end-to-end reproduction of a 2026 adaptive-jailbreak framework (arXiv:2505.23404) — prompt mutation, binary-tree encoding, dual-cipher output encryption, capability-classification probe — across three local models, plus an ablation and the adaptive defense the original paper names in its limitations but never evaluates.
- Independently replicated the judge-reliability finding: a self-judge was wrong in *both* directions on the same prompts; all verdicts re-checked by hand and the corrections committed alongside the originals.

ENGINEERING PROJECTS

Secure Guard — AI-driven incident response & SOC automation

github.com/secslayer/ai-soc

- Built a telemetry pipeline (Sysmon/Winlogbeat → Logstash → Elasticsearch) normalizing 45K+ events, with Python and SQL cleaning to extract analysis-ready threat datasets.
- Trained XGBoost and SVM classifiers with PCA-optimized cross-validation, reaching ~85% accuracy and balanced precision across highly varied malware datasets; Prophet forecasting converted statistical thresholds into concrete response playbooks.

GptGram — production-ready agentic AI platform

github.com/secslayer/Gptgram-production-ready

- Deployed an orchestration backend on AWS — REST APIs and JSON payloads over distributed Celery queues — handling high-concurrency LLM inference for multi-agent reasoning.
- Engineered semantic RAG pipelines on Qdrant, persisting conversational state to S3, under enforced Git/CI-CD and Agile practice.

Ggram — collaborative agentic AI platform

github.com/secslayer/Ggram

- Designed an agentic platform using graph neural networks to autonomously solve ML routing tasks. Earlier PoC: a scikit-learn NLP chatbot that cut simulated SOC false positives by ~30% via ensemble predictive models.

TECHNICAL SKILLS

- **AI security:** jailbreak evaluation, LLM red-teaming, adversarial ML, judge/eval validity, prompt injection, activation steering and linear probes.
- **Agentic systems:** multi-agent orchestration, RAG pipelines, Qdrant, LangChain, LLM inference at concurrency.
- **Defensive security:** detection engineering, SIEM (ELK, Splunk), incident response, threat hunting, MITRE ATT&CK.
- **ML & data:** PyTorch, scikit-learn, XGBoost, Pandas, NumPy, SQL, Databricks, Power BI.
- **Cloud & engineering:** AWS (EC2, S3), Docker, Git/CI-CD, REST APIs, JSON, Agile/Scrum.

CERTIFICATIONS

- **CompTIA CySA+** (Cybersecurity Analyst, CS0-003) — [verify](#)
- **Mathematics for Machine Learning**, Imperial College London via Coursera — linear algebra, multivariate calculus, PCA — [verify](#)
- **SOC Analyst**, TryHackMe — [certificate](#)
- **AWS Certified Generative AI Developer (AIP-C01)** — preparation course completed (exam not yet sat) — [certificate](#)
- **HTB Certified Offensive AI Expert (COAE)** — AI red-teamer path, in progress.

EDUCATION

TKM Institute of Technology — B.Tech, Computer Science & Engineering

Graduated 2025 | CGPA 7.56/10.00

Also: Kaggle competitor (ARC Prize 2026, ARC-AGI-3); high rankings in Hack The Box and CTF competitions; ACM chapter participation; TryHackMe and HTB workshops on detection engineering, incident response, and MITRE ATT&CK.